Camouflage Unlearning: Fast and Surgical Machine Unlearning for Biometric Systems

***Abstract*—Machine unlearning for privacy-critical biometric systems such as fingerprint, iris, and face recognition remains underexplored, despite the growing need for efficient solutions under General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) mandates that require data erasure requests to be supported throughout a model's deployment lifecycle. Existing methods predominantly target classification and generative tasks, and rely on many-to-one feature collapse that forces all forget classes toward a single distant point in feature space. This increases convergence time and computational cost, making them impractical for resource-constrained edge deployments where biometric models face frequent sequential removal requests with minimal replay data. Additionally, existing methods have been benchmarked almost exclusively on ResNet or ViT backbones for CIFAR and ImageNet, leaving their effectiveness on biometric backbones largely untested. In this work, we propose Camouflage Unlearning (CU), a cycle-robust framework that maps forget classes toward nearby retain-class boundaries rather than collapsing to a single point, thereby reducing migration distance. CU achieves 3–4$\times$ faster convergence through retain-dominance constraints and localized regularization that performs surgical updates in specific transformer blocks, while maintaining competitive retain accuracy. Evaluation across four domains, classification, iris, fingerprint, and face recognition demonstrates competitive performance and privacy behavior (MIA $\approx$ 0.5), supporting CU's effectiveness for biometric authentication systems requiring regulatory compliance.**

***Index Terms*—Biometric recognition, data privacy, face recognition, GDPR, machine unlearning, vision transformers**

I. Introduction

MACHINE unlearning (MU) has emerged as a critical research area, addressing the imperative to prevent unauthorized data exploitation, eliminate harmful model behaviors, and safeguard user privacy [1]. A natural consideration follows: what drives the necessity for unlearning? The proliferation of AI misuse ranging from phishing generation and model inversion attacks to perpetuating misinformation has prompted regulatory frameworks [2], [3], [4]. Frameworks like the General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA) [5], [6] mandate data erasure capabilities. These regulations require model service providers to erase user data from learned representations upon request. This establishes both legal and ethical obligations for responsible AI deployment. Crucially, users possess the right to request data removal at any time. This transforms unlearning from a one time operation into a continuous process throughout a model's deployment lifecycle. Per-request efficiency, therefore, becomes a first-class concern.

Existing machine unlearning methods predominantly target image classification [7] and generative tasks [8], with limited exploration of privacy-critical biometric systems. While face recognition [9], [10], [11] has received modest attention, fingerprint and iris recognition remains substantially less studied. This gap persists despite their sensitive nature and widespread deployment in authentication systems. Contemporary approaches rely primarily on KL divergence maximization [12] or gradient ascent [13]. These methods implement many-to-one mappings that force forgotten class embeddings toward a single distant point in feature space. Alternative strategies include saliency-based selection [14], geometric boundary manipulation [15], perturbation-based forgetting [16], and meta-learning optimization [17], [18]. A key observation across all these methods is that every forget sample is driven toward a single collapse point in the decision space. Even when a sample already lies far from the retain manifold, it must still traverse the full distance to this distant target. This increases convergence time and GPU cost, as observed in Table I. This many-to-one collapse behavior is visualized and ablated in detail in Section III-B. Concretely, GS-LoRA [13] (group-sparse LoRA, slow ascent), SCRUB [12] (bidirectional logits, unstable KL), SalUn [14] (weight saliency, reversible labels), Boundary Unlearning [15] (shadow-node pruning, slow), Forget Vectors [16] (noise tuning, invertible), MCU [18] (Bézier curves, dual optimization), LTU [17] (meta-learning, computationally expensive), DELETE [19] and LoTUS [20] (frozen teacher, double memory), and MUNBa [21] (Nash bargaining, per-step overhead) each trade convergence speed, memory, or reversibility, and few explicitly target biometric edge deployment or sequential forget cycles.

Biometric recognition differs from general classification in ways that directly affect unlearning feasibility. Biometric systems deploy on resource-constrained edge devices. Removal requests arrive sequentially and frequently over the deployment lifecycle. This transforms unlearning into a recurring operational cost rather than a one-time procedure. Under this workload, existing many-to-one methods become prohibitive as per request compute scales poorly with request frequency. Compounding this, existing unlearning methods have been designed and benchmarked almost exclusively on ResNet [22] or ViT backbones [23] for CIFAR [24] and ImageNet [25]. Their transferability to biometric backbones such as Face Transformer [26] (sliding-patch ViT face recognition), LAM [27] (ResNet-50, lightweight attention), JIPNet [28] (CNN-Transformer joint pose-identity), SSLDMNet-Iris [29] (CNN-GRU, Fisher discriminant), IrisFormer [30] (rotation-invariant ViT matching), PFSimNet [31] (partial fingerprint similarity network), and MEM-FR [32] (optical, privacy-preserving recognition) remains untested, despite these architectures being the standard in deployed authentication systems. What biometric unlearning requires instead is a cycle-robust method validated directly on biometric backbones.

To address these challenges, we propose Camouflage Unlearning (CU), a novel framework that achieves 3–4× faster convergence while maintaining robust performance. Unlike existing many-to-one approaches, CU pushes forget-class embeddings toward the nearest retained class boundaries, effectively reducing migration distances. To perform surgical modifications and minimize collateral damage to retain classes, CU employs localized regularization concentrated in specific blocks of the neural network for additional details, please refer to Section II.

Our main contributions are summarized as follows:

1) To the best of our knowledge, this is the among the earliest comprehensive studies of machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition.
2) We introduce Camouflage Unlearning, which achieves 3–4× faster convergence compared to SoTA unlearning methods while maintaining competitive retention accuracy and privacy guarantees.
3) We conduct extensive empirical evaluation across multiple biometric and classification domains, demonstrating that CU concentrates updates in specific network blocks and remains effective under data-scarce replay conditions.

II. Problem Setting and Methodology

A. Problem Setting

Let $\mathcal{M}$ be a model pre-trained on $D = D_r^{full} \cup D_f$ ($D_r^{full} \cap D_f = \emptyset$), representing a mapping $f_{\mathcal{M}} : \mathcal{X}_D \to \mathcal{Y}_D$. Storing D_r^{full} is prohibited under GDPR/CCPA, and full retraining is costly, so we retain only a small replay buffer $D_r \subset D_r^{full}$, with $|D_r|$ limited to at most 10% of $|D_r^{full}|$[1]. Before unlearning, $\mathcal{M}$ correctly maps both D_f and D_r. Applying an unlearning algorithm $\mathcal{F}$ (Section II-B) yields $\mathcal{M}' = \mathcal{F}(\mathcal{M}, D_f, D_r)$, under which the forget mapping should no longer hold while the retain mapping persists:

$$f_{\mathcal{M}} : \mathcal{X}_{D_f} \to \mathcal{Y}_{D_f} \,;\, \mathcal{X}_{D_r} \to \mathcal{Y}_{D_r} \implies f_{\mathcal{M}'} : \mathcal{X}_{D_f} \not\to \mathcal{Y}_{D_f} \,;\, \mathcal{X}_{D_r} \to \mathcal{Y}_{D_r} \tag{1}$$

where $\not\to$ denotes a forgotten mapping and $\to$ a retained one. We define successful unlearning as

- Forgetting: $Acc(\mathcal{M}', D_f) \leq \frac{1}{C}$,
- Retention: $Acc(\mathcal{M}', D_r) \approx Acc(Oracle, D_r)$

where C is the number of classes and Oracle denotes a model trained directly on target classes only.

B. Camouflage Unlearning

CU relocates forget-class embeddings to retain-class boundaries via direct centroid mapping. Unlike existing many-to-one approaches that collapse all forget classes to a single distant point, CU implements many-to-many distribution, reducing migration distance and accelerating convergence.

We compute class centroids by averaging embeddings, $c_k = \frac{1}{|D_k|}\sum_{x \in D_k} emb(x)$, yielding retain centroids $\{c_{r_1}, \ldots, c_{r_K}\}$ and forget centroids $\{c_{f_1}, \ldots, c_{f_M}\}$, where K and M are the numbers of retain and forget classes. Each forget centroid c_{f_i} is assigned its nearest retain centroid as target:

$$\mathbf{t}_i = c_{r_{j^*}}, \quad j^* = \arg\min_j \|c_{f_i} - c_{r_j}\|_2 \tag{2}$$

directly mapping forget embeddings into retain decision boundaries and minimizing migration distance. We pull forget embeddings toward their assigned targets:

$$\mathcal{L}_f = \frac{1}{|D_f|} \sum_{(\mathbf{x},y) \in D_f} \|emb(\mathbf{x}) - \mathbf{t}_y\|^2 \tag{3}$$

which collapses forget classes into retain boundaries but risks corrupting retain decision regions and prolonging convergence. We therefore enforce that retain logits dominate forget logits by margin $\gamma = 2.0$:

$$\mathcal{L}_{dom} = ReLU\left(\max_{i \in F} z_i + \gamma - \max_{j \in R} z_j\right) \tag{4}$$

where z_i are logits and F, R denote forget and retain class indices; this accelerates forget-class suppression while preserving retain accuracy. Retain-class knowledge is preserved through classification and embedding stability:

$$\mathcal{L}_{ret} = \lambda_{ce} \cdot CE(\mathbf{z}_r, y_r) + \lambda_{emb} \cdot \frac{1}{|D_r|} \sum_{(\mathbf{x},y) \in D_r} \|emb(\mathbf{x}) - c_y\|^2 \tag{5}$$

where $\mathbf{z}_r$ are retain logits, y_r are labels, and c_y are original retain centroids. Since comprehensive parameter updates risk instability under limited replay data, inspired by group sparse selection [13], we apply localized regularization that selectively updates fewer transformer blocks:

$$\mathcal{L}_{local} = \lambda_{local} \sum_{g=1}^{G} \|\theta_g\|_2 \tag{6}$$

where θ_g represents parameters in block g and G is the total number of groups. Each θ_g is initialized at zero and tracks the *change* accumulated in block g relative to the pretrained weights, so $\mathcal{L}_{local}$ penalizes the magnitude of updates rather than absolute parameter magnitude. This group-Lasso-inspired approach concentrates updates in specific blocks while leaving others unchanged, preserving retained knowledge under constrained replay. The total loss integrates all components:

$$\mathcal{L}_{total} = \mathcal{L}_f + \lambda_{dom}\mathcal{L}_{dom} + \mathcal{L}_{ret} + \mathcal{L}_{local} \tag{7}$$

III. Performance Evaluation

A. Experimental Setup

Datasets, Backbones, and Baselines: We evaluate on CIFAR-100 [24] (DeiT-Tiny [33]), CASIA-Iris [34] (LAM [27]), Sokoto Coventry fingerprints [35] (JIPNet [28]), and CASIA-Face100 [36] (Face Transformer [26]), comparing CU against SCRUB [12], SalUn [14], GS-LoRA [13], LTU [17], MCU [18], and Forget Vectors [16], plus an Oracle trained only on $D_r^{\mathbf{full}}$ as the upper bound.

Evaluation Metrics: We report Acc_f, Acc_r, MIA (label-only attack [37], ≈ 0.5 = safe), KL divergence to Oracle, RTE, and H-Mean [38]. Success requires $Acc_f \approx 0$, $Acc_r \approx$ Oracle,

[1]From this point, D_r denotes the replay buffer unless specified.

Method	Classification						Iris recognition					
	$Acc_f\downarrow$	$Acc_r\uparrow$	$H\text{-}Mean\uparrow$	KL$\downarrow$	MIA$\downarrow$	RTE$\downarrow$	$Acc_f\downarrow$	$Acc_r\uparrow$	$H\text{-}Mean\uparrow$	KL$\downarrow$	MIA$\downarrow$	RTE$\downarrow$
Oracle	–	77.36	–	–	–	–	–	99.03	–	–	–	–
GS-LoRA [13]	0.11	71.32	74.18	0.73	0.56	153.2	0.00	93.42	96.18	0.75	0.69	225.7
SalUn [14]	0.37	72.46	74.63	1.32	0.63	225.6	0.37	92.37	95.44	0.79	0.59	427.2
SCRUB [12]	0.58	73.11	74.90	1.89	0.64	219.3	0.43	95.58	97.07	0.83	0.63	501.8
FV [16]	0.00	74.43	75.89	0.87	0.55	189.5	0.00	96.27	97.63	1.13	0.57	383.1
MCU [18]	0.42	74.43	75.66	0.87	0.59	175.5	0.00	98.87	98.95	0.99	0.63	437.2
LTU [17]	0.00	76.32	76.83	0.75	0.57	239.3	0.73	95.43	96.85	1.25	0.55	329.4
CU	0.47	74.46	75.65	1.27	0.56	55.5	0.06	97.89	98.43	1.01	0.53	116.1

Method	Fingerprint recognition						Face recognition					
	$Acc_f\downarrow$	$Acc_r\uparrow$	$H\text{-}Mean\uparrow$	KL$\downarrow$	MIA$\downarrow$	RTE$\downarrow$	$Acc_f\downarrow$	$Acc_r\uparrow$	$H\text{-}Mean\uparrow$	KL$\downarrow$	MIA$\downarrow$	RTE$\downarrow$
Oracle	–	90.82	–	–	–	–	–	95.12	–	–	–	–
GS-LoRA [13]	0.00	83.42	86.98	1.13	0.65	155.3	0.57	93.25	93.90	1.13	0.53	284.4
SalUn [14]	0.10	82.32	86.24	0.75	0.63	202.2	0.00	90.15	92.56	1.27	0.59	417.2
SCRUB [12]	0.00	83.27	86.85	2.36	0.57	175.3	0.42	93.27	93.98	2.87	0.63	389.5
FV [16]	0.30	84.44	87.38	1.27	0.54	160.7	0.20	92.27	93.58	1.20	0.60	380.3
MCU [18]	0.00	80.32	85.24	0.99	0.59	220.5	0.00	90.12	92.54	0.89	0.57	403.8
LTU [17]	0.00	87.72	89.24	1.13	0.54	237.4	0.00	92.58	93.84	0.93	0.55	430.2
CU	0.00	87.23	88.98	1.05	0.55	55.32	0.00	93.98	94.55	1.67	0.51	175.3

TABLE I

COMPREHENSIVE PERFORMANCE EVALUATION ACROSS FOUR RECOGNITION DOMAINS. PROPOSED CU ACHIEVES 3–4$\times$ SPEEDUP IN RUNTIME EFFICIENCY (RTE) WHILE MAINTAINING COMPETITIVE FORGETTING, RETENTION, AND PRIVACY METRICS .

low KL/RTE. FAR@$\{0.3, 0.4, 0.5\}$/EER (cosine similarity, forget probes vs. retain gallery) further verify open-set identity confusion (Section III-D).

B. Main Results

Table I compares all methods across four tasks; green/blue mark best/second-best.

Results: All methods forget near-perfectly ($Acc_f < 1\%$). CU ranks first or second in Acc_r and H-Mean across all tasks (LTU and MCU the closest competitors), and achieves MIA ≈ 0.5 despite GS-LoRA's lower KL, expected since CU deliberately reshapes forget logits. Most critically, CU delivers a 3–4$\times$ RTE speedup over every baseline.

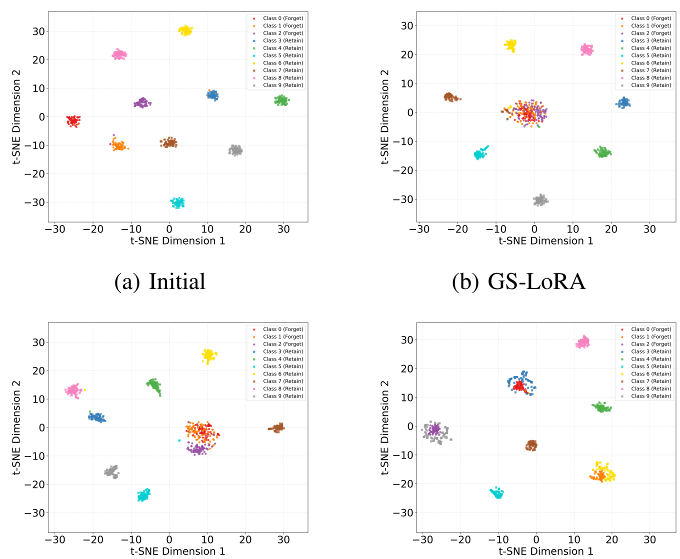
Geometric Analysis: Figure 1 visualizes embedding-space t-SNE. GS-LoRA [13], SCRUB [12], and Boundary Unlearning [15] collapse forget classes to one distant point; CU instead migrates them to the nearest retain boundary, explaining its RTE gains in Table I.

C. Ablation Studies

Replay Buffer Size: We investigate the minimal replay buffer size $|D_\mathbf{r}|$ for effective unlearning on CIFAR-100. Table II shows that a 0.1 ratio optimally balances retention accuracy and efficiency.

Retain Dominance Constraint: Without this constraint, we found that the forget accuracy exhibits a prolonged tail, delaying convergence. Table III demonstrates that applying the constraint reduces Acc_f from 9.97% to 0.39%, achieving faster unlearning with minimal retention loss.

Localized Parameter Regularization: This regularization selectively modifies fewer blocks, enabling surgical edits that preserve model capacity. On CIFAR-100 classification, Fig-



(a) Initial (b) GS-LoRA

(c) Boundary (d) CU (Ours)

Fig. 1. t-SNE visualization. Baselines collapse to single point while proposed CU method distributes to nearest boundaries thus achieving faster convergence.

Buffer Ratio	Speed	Acc_f	Acc_r	$H\text{-}Mean$
1.0	1$\times$	1.33	74.39	2.62
0.5	2$\times$	1.18	73.00	2.32
0.3	3.3$\times$	1.33	74.55	2.61
0.1	10$\times$	1.30	74.52	2.56

TABLE II

IMPACT OF REPLAY BUFFER SIZE ON UNLEARNING PERFORMANCE. A RATIO OF 0.1 OPTIMALLY BALANCES RETENTION AND EFFICIENCY.

Method	Acc_f	Acc_r
Without Constraint	9.97	78.12
With Constraint	0.39	77.06

TABLE III
ABLATION ON RETAIN DOMINANCE CONSTRAINT. THE CONSTRAINT ACCELERATES UNLEARNING BY ELIMINATING THE LONG TAIL CONVERGENCE ISSUE.

ure 2 shows concentrated updates in specific blocks, while Table IV demonstrates reducing total ℓ_2 norm from 39.05 to 10.09 with minimal retention loss.

Method	Acc_f	Acc_r	Total ℓ_2 Norm
Without Localized Reg	0.00	77.24	39.05
With Localized Reg	0.00	74.76	10.09

TABLE IV
LOCALIZED REGULARIZATION REDUCES TOTAL ℓ_2 NORM WHILE MAINTAINING EFFECTIVE UNLEARNING.

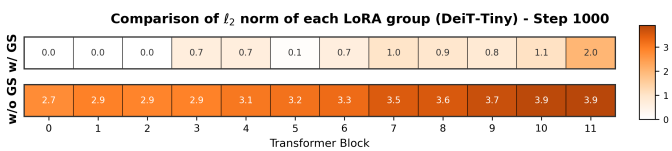


Fig. 2. ℓ_2 norm per block. Localized regularization concentrates updates in specific blocks.

Scalability: We apply CU to five ViT backbones (DeiT-Tiny to ViT-Large). Table V shows consistent forgetting and retention across both lightweight and large-scale architectures.

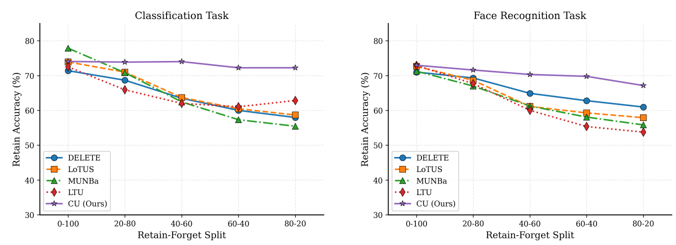
Model	Before Unlearning		After Unlearning	
	Acc_f	Acc_r	$Acc_f \downarrow$	$Acc_r \uparrow$
DeiT-Tiny	75.25	75.67	0.40	74.46
DeiT-Small	74.86	74.01	1.35	73.89
DeiT-Base	74.32	74.15	0.87	72.58
ViT-Base	72.58	73.21	2.35	70.35
ViT-Large	75.35	78.91	0.10	75.61

TABLE V
EFFECT OF VIT BACKBONE SCALE ON UNLEARNING PERFORMANCE.

Sequential Removal Workload: We pilot CU against DELETE [19], LoTUS [20], MUNBa [21], and LTU [17] across successive forget cycles on CIFAR-100 [24] and CASIA-Face100 [36]. Figure 3 shows baselines degrade progressively while CU retains stable accuracy, confirming robustness under continual removal.

D. Attack-Based Verification

Table VI reports FAR/EER of forget-class probes matched against the retain gallery on CASIA-Iris. CU tracks the Oracle closely (FAR@0.5 3.83% vs. 3.10%; EER 24.99% vs. 21.69%), and both rise similarly over *Before* unlearning



(a) CIFAR-100 (b) Face Recognition

Fig. 3. Retain accuracy across sequential unlearning cycles. Existing methods degrade progressively while CU maintains stable performance on both benchmarks.

(1.64%), indicating that the small increase in open-set false accepts stems from removing the identities rather than from CU specifically.

Model	FAR@0.3	FAR@0.4	FAR@0.5	EER
Before	11.38%	4.84%	1.64%	13.33%
CU	17.97%	8.52%	3.83%	21.99%
Oracle	16.10%	7.79%	3.10%	21.69%

TABLE VI
FAR/EER OF FORGET-CLASS PROBES AGAINST THE RETAIN GALLERY ON CASIA-IRIS.

Attention Maps: We visualize attention on forget/retain CASIA-Face100 [36] samples before/after CU. Figure 4 shows the base model attends to facial regions for both; retrained and CU models shift attention to peripheral areas *only for forget samples*, confirming CU surgically removes identity-specific features while preserving retain ones.

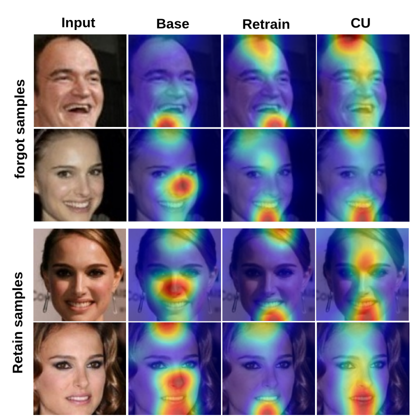


Fig. 4. Attention maps on forget and retain samples. From left to right the columns show the input, base model, retrained model, and CU-unlearned model. Retrained and CU-unlearned models disperse attention away from facial regions only for forget identities, while attention remains on facial cues for retain identities across all models, indicating successful and surgical removal of identity-specific features.

Relearning Attacks: We test whether CU's forgetting is genuine or reversible via a relearning attack: fine-tuning a CU-unlearned model (50 classes) and a head-masked baseline on forgotten classes, tracking recovery. Figure 5 shows

CU recovers gradually versus head-masking's rapid recovery, confirming genuine backbone-level forgetting.

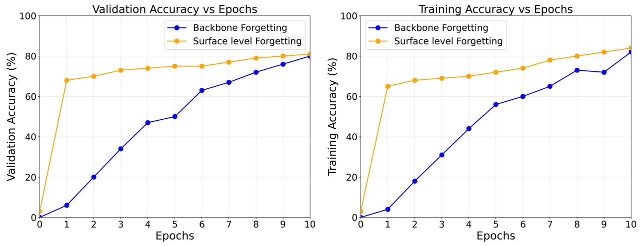


Fig. 5. Backbone vs surface-level unlearning. CU recovers gradually (genuine forgetting) while head-masking recovers rapidly (superficial).

IV. Results Discussion

A. On the Necessity of the Retain Buffer

A natural concern for Camouflage Unlearning is whether maintaining the replay buffer $D_r \subset D_r^{\text{full}}$ contradicts GDPR and CCPA mandates on data erasure. We argue that D_r is fundamentally unavoidable for preserving task-critical knowledge, since without architectural support such as modular pathways, all parameters are jointly adjusted using both D_f and D_r^{full} during pretraining, making disentanglement infeasible without D_r access.

Attempts to bypass D_r are limited. Impair and repair methods [39] still require D_r for repair, while source-free unlearning [40] estimates retain data Hessians from D_f alone but is restricted to convex losses and linear classifiers, making it inapplicable to transformers and ResNets. All established non-convex methods critically rely on D_r, whether through Fisher information or saliency [41], [14], explicit retention losses [12], [13], or continual unlearning mechanisms such as mnemonic codes, task embeddings, experience replay, and stability regularizers [38], [42], [43], [44], [45].

Generative tasks offer exceptions such as FLAT [46] and ESD [47], which avoid D_r via regularizers or teacher-generated samples, but few practical approaches eliminate D_r for non-convex deep learning without sacrificing retention. Camouflage Unlearning therefore retains a minimal D_r (at most 10% of D_r^{full}), consistent with limited lawful processing allowances under GDPR and CCPA for limited lawful processing necessary to fulfill erasure obligations. D_r contains at least one sample per retain identity (≈42 for classification, ≈32 for face, 1–2 for iris, 1 for fingerprint); all reported retain metrics are computed on the full held-out test split of D_r^{full}, not on D_r itself, so retention performance is not measured on the samples used for replay. Furthermore, D_r holds only consenting retain identities: a withdrawal request is handled as a forget request, and the buffer is refilled from the remaining consenting pool, keeping the replay mechanism consistent with limited lawful processing under GDPR and CCPA.

V. Conclusion

To our knowledge, we are among the earliest to address machine unlearning for privacy-critical biometric systems, including fingerprint, iris, and face recognition. We observed that existing methods suffer from prolonged convergence due to many-to-one embedding collapse, which is particularly inefficient for biometric models experiencing frequent unlearning requests on edge devices. To address this, we propose Camouflage Unlearning, achieving up to 4× speedup through many-to-many distribution and surgical parameter modifications while delivering competitive forgetting, retention, and privacy performance comparable to state-of-the-art methods. Future work includes extending Camouflage Unlearning to continual forgetting scenarios in pretrained biometric models.

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